Homework #3: Gaussians & Uncertainty Prop

GS-Q1) N-dim Gaussian. double precision numbs used to represent scalars (8 bytes each)

(a) memory to store state vector $= 8N$

(b) mem to store covariance

$N \times N$ covariance matrix $\longrightarrow 8N^2$

only need upper triangle $8 \cdot \left(\frac{N(N+1)}{2}\right)$

$4N^2 + 4N$

GS-Q2) $p(x,y) \sim \mathcal{N}(\vec{\mu}, \Sigma)$

$$= \frac{1}{|2\pi\Sigma|^{1/2}} e^{-\frac{1}{2}((\vec{x}-\vec{\mu})^T \Sigma^{-1} (\vec{x}-\vec{\mu}))}$$

$\vec{Y} = [x,y]^T$, $\mu = [\mu_x, \mu_y]^T$ $\Sigma = \begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{bmatrix}$

show cond only results in

$$p(x|y) = \mathcal{N}(\mu_*, \sigma_*^2) = \frac{1}{(2\pi\sigma_*^2)^{1/2}} e^{-\frac{1}{2}\frac{(x-\mu_*)^2}{\sigma_*^2}}$$

where $\mu_* = \mu_x + \frac{\sigma_{xy}}{\sigma_y}(y-\mu_y)$ $+ \sigma_*^2 = \sigma_x^2 - \frac{\sigma_{xy}^2}{\sigma_y^2}$

$$p(x|y) = \frac{p(x,y)}{p(y)}$$

$$p(x,y) = \frac{1}{|2\pi\begin{bmatrix} \sigma_x^2 & \sigma_{xy} \\ \sigma_{xy} & \sigma_y^2 \end{bmatrix}|^{1/2}} e^{-\frac{1}{2}\left(\begin{bmatrix} x \\ y \end{bmatrix} - \begin{bmatrix} \mu_x \\ \mu_y \end{bmatrix}\right)^T \begin{bmatrix} \sigma_y^2 & -\sigma_{xy} \\ -\sigma_{xy} & \sigma_x^2 \end{bmatrix} \left(\begin{bmatrix} x-\mu_x \\ y-\mu_y \end{bmatrix}\right) \cdot \frac{1}{|\Sigma|}}$$

$$\Sigma^{-1} = \frac{1}{|\Sigma|}\begin{bmatrix} \sigma_y^2 & -\sigma_{xy} \\ -\sigma_{xy} & \sigma_x^2 \end{bmatrix} \qquad = \frac{1}{|2\pi\Sigma|^{1/2}} e^{-\frac{1}{2}\begin{bmatrix} x-\mu_x, y-\mu_y \end{bmatrix} \begin{bmatrix} \sigma_y^2 & -\sigma_{xy} \\ -\sigma_{xy} & \sigma_x^2 \end{bmatrix}}$$

$$= \frac{1}{|2\pi\Sigma|^{1/2}} e^{-\frac{1}{2}\left[\frac{1}{|\Sigma|}, \sigma_y^2 u^2 - 2\sigma_{xy}uv + \sigma_x^2 v^2\right]} \qquad u = x-\mu_x \quad v = y-\mu_y$$

cond only $\rightarrow$ v is known const

$$= \frac{1}{|2\pi\Sigma|^{1/2}} e^{-\frac{1}{2}\left(\frac{1}{|\Sigma|}\right)\left(\sigma_y u^2 - \frac{2\sigma_{xy}uv}{\sigma_y^2}\cdot\sigma_y^2\right\} + \sigma_x v^2}$$

$$= \frac{1}{|2\pi\Sigma|^{1/2}} e^{-\frac{1}{2}\left(\frac{\sigma_y^2}{|\Sigma|}\left(u - \frac{\sigma_{xy}}{\sigma_y^2}v\right)^2 - \frac{\sigma_{xy}^2}{\sigma_y^2|\Sigma|}v^2 + \frac{\sigma_x^2}{|\Sigma|}v^2\right)}$$

$$\frac{-\sigma_{xy}^2 + \sigma_x^2\sigma_y^2}{\sigma_y^2|\Sigma|}v^2 = \frac{v^2}{\sigma_y^2}$$

$$= \frac{1}{|2\pi\Sigma|^{1/2}} e^{-\frac{1}{2}\left(\frac{\sigma_y^2}{|\Sigma|}\left(u - \frac{\sigma_{xy}}{\sigma_y^2}v\right)^2 + \frac{v^2}{\sigma_y^2}\right)}$$

$$\frac{1}{|2\pi\Sigma|^{1/2}} e^{-\frac{1}{2}\left(\frac{(y-\mu_y)^2}{\sigma_y^2}\right)}$$

GS-Q3) $x \sim \mathcal{N}(\mu_x, \Sigma_x)$, $y = Ax+b$ also gauss.

$y = Ax+b$ $\quad \mu_y = E[y] = E[Ax+b]$

$= AE[x]+b$

$= A\mu_x + b$

$\Sigma_y = E[(y-\mu_y)(y-\mu_y)^T]$

$= E[(Ax+b-A\mu_x-b)(Ax+b-A\mu_x-b)^T]$

$= E[(Ax-A\mu_x)(Ax-A\mu_x)]$

$= E[A(x-\mu_x)(x-\mu_x)^T A^T]$

$= A\Sigma_x A^T$

UP-Q0) all UP questions in code